

# Software Requirements Specification

## *Real-Time QR Lead Generation and Company Profile System*

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## 1. System Overview

The system is a customizable QR-based company profile and real-time lead generation platform. It allows different company profiles to be created from the same reusable system. The company name, logo, video, text content, services, colors, and public contact information are replaceable for each profile. However, the lead receiver/notification destination is a locked system setting and shall not be changed by company-level users.

When a visitor scans the QR code, they are directed to the company's landing page. After watching the video and viewing the company information, the visitor may submit their contact details. Once submitted, the lead information is saved and sent in real time to the fixed lead receiver configured by the system owner/super admin.

## 2. Definitions and Conventions

### 2.1 Modal Verb Conventions

Throughout this document, the following modal verbs are used as defined below:

| Verb  | Meaning                                            |
|-------|----------------------------------------------------|
| shall | Mandatory requirement. Non-compliance is a defect. |

|        |                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------|
| should | Recommended. Non-compliance requires documented justification.                                   |
| may    | Optional. Implementation is at the discretion of the development team or based on configuration. |

## 2.2 Key Terms

| Term                | Definition                                                                                                                                                                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead                | A visitor who has submitted the contact form on a company landing page.                                                                                                                                                                    |
| Company Profile     | The replaceable set of company information, media, branding, and public landing page content associated with one company/profile in the platform.                                                                                          |
| QR Code             | A machine-readable two-dimensional barcode that encodes a URL to a company landing page.                                                                                                                                                   |
| Landing Page        | The public-facing web page a visitor sees after scanning the QR code.                                                                                                                                                                      |
| Admin               | A system administrator with full platform access.                                                                                                                                                                                          |
| Company Rep         | An authorized representative of a company with access limited to their company's data.                                                                                                                                                     |
| Real-time           | Delivery within 5 seconds of the triggering event under normal operating conditions.                                                                                                                                                       |
| Notification        | An out-of-band alert (email, Telegram, etc.) sent to the fixed lead receiver when a new lead is submitted.                                                                                                                                 |
| Fixed Lead Receiver | The person, email address, Telegram chat, or internal account that receives submitted lead/customer information. This receiver is controlled only by the system owner/super admin and is not replaceable by company-level content changes. |

## 3. Main User Flow

The system shall follow this primary process:

**Scan QR -> Watch short video -> View company info -> Fill contact form -> Submit -> Lead saved to database -> Fixed receiver notified in real time**

The system shall also support these secondary flows:

- Admin login → Manage companies → View leads → Export leads
- Company rep login → View own leads → Receive notifications → Follow up

## 4. Main Purpose

The main purpose of the system is to help companies collect potential customer information instantly through QR-based marketing.

The system shall allow companies to:

- Share company introduction videos.
- Present company information and services.
- Collect visitor contact information through a public form.

- Receive leads immediately after form submission through the fixed lead receiver.
- Follow up with interested visitors quickly.

## 5. Target Users and Roles

### 5.1 System Admin

The system admin manages the entire platform. There may be one or more admin accounts.

The admin shall be able to:

- Log in and log out securely.
- Add, edit, and delete company profiles.
- Upload or update video URLs for any company.
- Generate and view QR codes for any company.
- Activate or deactivate any company profile.
- View all leads across all companies.
- Search and filter leads by company, date range, and status.
- Export leads to CSV format.
- Manage admin account credentials.
- View the system audit log.
- Configure the fixed lead receiver/notification destination. Company-level users shall not be able to change this receiver.

### 5.2 Company Representative

The company representative manages leads for their own company only. They cannot access data belonging to other companies.

The company rep shall be able to:

- Log in and log out securely.
- View leads submitted to their own company.
- View real-time leads if permission is granted. Lead notifications shall go to the fixed receiver unless the super admin assigns the representative as that receiver.
- Mark leads with a follow-up status (if enabled by admin).

The company rep may be able to (if permission is granted by admin):

- Update their company's replaceable content, public contact details, and video URL. They shall not update the fixed lead receiver unless granted super admin-level permission.
- View their company's QR code and copy its link.

The company rep shall not be able to:

- Access any other company's leads or profile.
- Activate or deactivate their own company profile.
- Export leads unless specifically granted permission.
- Access admin functions, other company rep accounts, or fixed receiver configuration.

### 5.3 Public Visitor / Potential Customer

The visitor is an unauthenticated user who has scanned a QR code. No account is required.

The visitor shall be able to:

- View the company landing page on any mobile or desktop browser.
- Watch the embedded video.
- Read company information and services.
- Submit the lead capture form.

- Use direct contact buttons (phone, email, WhatsApp, website).

## 6. Functional Requirements

### 6.1 Authentication and Session Management

**FR-AUTH-01:** The system shall require login for all admin and company rep dashboard access. The landing page shall be publicly accessible without authentication.

**FR-AUTH-02:** The login form shall require an email address and password.

**FR-AUTH-03:** After 5 consecutive failed login attempts, the account shall be locked for 15 minutes. The user shall be shown a message indicating the lockout and its duration.

**FR-AUTH-04:** The system shall provide a password reset flow via email for locked or forgotten credentials.

**FR-AUTH-05:** Authenticated sessions shall expire after 60 minutes of inactivity. The user shall be redirected to the login page and shown a session-expired message.

**FR-AUTH-06:** The system shall not allow concurrent login sessions for the same account from different devices without explicit session management (such as a "log out other devices" option or a warning).

**FR-AUTH-07:** Passwords shall be stored using a strong one-way hashing algorithm (bcrypt or Argon2). Passwords shall never be stored in plain text or reversibly encrypted form.

### 6.2 Company Profile Management

**FR-CPM-01:** The system shall allow the admin to create, edit, replace, and deactivate company profiles.

**FR-CPM-01A:** The system shall support reusable company profiles where the company name, logo, brand color, video URL, descriptions, services/products, public contact links, website, social links, and address can be changed without changing the system structure or QR code URL.

**FR-CPM-01B:** Replacing company content shall not replace or modify the fixed lead receiver/notification destination.

**FR-CPM-02:** Each company profile shall include the following fields:

| Field                     | Type                                      | Required |
|---------------------------|-------------------------------------------|----------|
| Company name              | Text, max 120 characters                  | Yes      |
| Logo                      | Image file (JPEG/PNG/WebP, max 2 MB)      | Yes      |
| Brand primary color       | Hex color code                            | No       |
| Short description         | Text, max 500 characters                  | No       |
| Services/products         | Text, max 1,000 characters                | No       |
| Video URL                 | Valid URL (YouTube, Vimeo, or Cloudinary) | No       |
| Public contact name/title | Text, max 100 characters                  | No       |
| Public contact email      | Valid email address                       | No       |
| Public contact phone      | Phone number with country code            | No       |
| WhatsApp link             | Valid URL or phone number                 | No       |
| Telegram link             | Valid URL or username                     | No       |
| Website URL               | Valid URL                                 | No       |
| Social media links        | Up to 5 URLs                              | No       |

|                         |                         |    |
|-------------------------|-------------------------|----|
|                         |                         |    |
| Address / location link | Text or Google Maps URL | No |

**FR-CPM-03:** Each company profile shall be assigned a unique system identifier at creation. This identifier shall not change and shall be the basis for the company's QR code URL.

**FR-CPM-04:** Admin shall be able to update replaceable company profile fields after creation without changing the company's unique identifier or QR code URL.

**FR-CPM-04A:** Updating company profile fields shall not update the fixed lead receiver unless the action is performed from the dedicated receiver configuration screen by a super admin.

**FR-CPM-05:** When a company profile is deactivated, its QR code link shall redirect to a configurable "not available" page. The message on that page shall be manageable by admin. It shall not show a 404 error.

**FR-CPM-06:** The system shall support a minimum of 200 active company profiles simultaneously.

### 6.3 QR Code Management

**FR-QR-01:** The system shall automatically generate a unique QR code for each company at the time the company profile is created.

**FR-QR-02:** Each QR code shall encode a permanent, unique URL in the format: [base\_domain]/c/[company\_identifier].

**FR-QR-03:** The QR code URL shall remain unchanged even if the company's name, video, public contact details, logo, or any other replaceable profile content is updated.

**FR-QR-03A:** QR code behaviour shall remain connected to the fixed lead receiver rule. Updating or replacing the company profile shall not change where submitted leads are sent.

**FR-QR-04:** The system shall generate QR codes at a minimum size of 300×300 pixels in PNG format, using error-correction level H (30% damage tolerance) to ensure readability when printed at small sizes.

**FR-QR-05:** Admin shall be able to download the QR code image for any company from the dashboard.

**FR-QR-06:** Admin shall be able to copy the QR code URL (the encoded link) to clipboard from the dashboard.

**FR-QR-07:** QR codes shall work on Android and iOS devices and on desktop browsers.

### 6.4 Dynamic Landing Page

**FR-LP-01:** Each active company shall have its own publicly accessible landing page at the URL encoded in the QR code.

**FR-LP-02:** The landing page shall display the following elements, in the following order, on mobile:

16. Company logo and name
17. Embedded short video (if a video URL is configured)
18. About / company description section
19. Services / products section
20. Contact details and direct contact buttons
21. Lead capture form

**FR-LP-03:** The landing page shall use the company's brand primary color as the accent color for buttons and headings, if that color has been configured.

**FR-LP-04:** If no video URL has been configured for a company, the video section shall be hidden. A placeholder image or the company logo shall appear in its place.

**FR-LP-05:** If a configured video URL is broken or unavailable at load time, the landing page shall still load fully. A friendly message shall be shown in the video area (e.g., "Video unavailable. Please contact us directly."). The rest of the page shall be unaffected.

**FR-LP-06:** The landing page shall load within 3 seconds on a 4G mobile connection under normal server load conditions.

**FR-LP-07:** The landing page shall be functional on Android, iOS, tablet, and desktop browsers.

## 6.5 Lead Capture Form

**FR-LCF-01:** The landing page shall include a lead capture form that visitors can submit without creating an account.

**FR-LCF-02:** The form shall collect the following fields:

| Field                        | Type                                               | Required |
|------------------------------|----------------------------------------------------|----------|
| Full name                    | Text, max 100 characters                           | Yes      |
| Phone number                 | Phone number with country code                     | Yes      |
| Email address                | Valid email format                                 | No       |
| Company / organization name  | Text, max 120 characters                           | No       |
| Interested service / product | Dropdown from company's service list, or free text | No       |
| Message / inquiry            | Textarea, max 1,000 characters                     | No       |
| Preferred contact method     | Radio: Phone / Email / WhatsApp                    | No       |

**FR-LCF-03:** The form shall validate all inputs on the client side before submission. The following validation rules shall apply:

- Full name: not empty, letters and common name characters only, min 2 characters.
- Phone number: not empty, valid phone format with country code.
- Email: if provided, must match standard email format (RFC 5322 compatible).
- Message: if provided, max 1,000 characters.

**FR-LCF-04:** The form shall validate inputs on the server side independently of client-side validation. Server-side validation shall be the authoritative check.

**FR-LCF-05:** The form submission button shall be disabled while a submission is in progress to prevent duplicate submissions.

**FR-LCF-06:** After a successful submission, the visitor shall see a thank-you confirmation message. The form shall not be re-submittable by refreshing the page (the system shall redirect or replace the form state).

**FR-LCF-07:** The system shall prevent the same visitor from submitting the form for the same company more than 3 times in a 24-hour window (based on IP address + phone number combination). If this limit is exceeded, the form shall display an appropriate message and shall not process the submission.

**FR-LCF-08:** The form shall include basic spam protection (CAPTCHA or similar) if the submission rate from a single IP exceeds 10 submissions in one hour.

## 6.6 Real-Time Lead Delivery

**FR-RLD-01:** When a visitor submits the form, the lead data shall be saved to the database before any notification is sent. Lead saving shall not depend on the success of notification delivery.

**FR-RLD-02:** The lead record saved to the database shall include:

- All submitted form fields.
- Timestamp of submission (UTC, ISO 8601 format).
- Company identifier.
- Visitor's IP address (hashed or anonymized for privacy compliance).
- Lead unique identifier.
- Submission status (new, viewed, followed-up, etc.).

**FR-RLD-03:** The admin and company rep dashboard shall update to show the new lead within 5 seconds of form submission, without manual page refresh.

**FR-RLD-04:** The system shall send a notification to the fixed lead receiver within 5 seconds of a successful form submission.

**FR-RLD-05:** Notification content shall include: company name, submission timestamp, visitor full name, phone number, email (if provided), and interested service (if provided).

**FR-RLD-06:** If notification delivery fails (email gateway error, Telegram API error, etc.), the system shall:

22. Retry delivery up to 3 times with exponential backoff (1s, 4s, 16s delays).
23. Log the failure with error details.
24. Mark the lead with a "notification failed" flag, visible to admin.
25. The lead data shall remain saved in the database regardless of notification failure.

**FR-RLD-07:** The system shall support at least email notification as the primary channel. Telegram notification shall be supported as a secondary channel.

**FR-RLD-08:** The fixed lead receiver shall be configured separately from replaceable company content. Company-level users shall not be able to edit, override, or replace this receiver from the company profile form.

**FR-RLD-09:** If a company profile is replaced with another company, all new leads from that QR landing page shall still be sent to the fixed lead receiver unless the super admin explicitly changes the receiver configuration.

## 6.7 Fixed Lead Receiver and Content Replacement Rule

**FR-FLR-01:** The system shall clearly separate replaceable company content from the fixed lead receiver configuration.

**FR-FLR-02:** Replaceable company content shall include company name, logo, video URL, description, services/products, public contact buttons, brand color, website, social links, and address/location.

**FR-FLR-03:** The fixed lead receiver shall include the destination where submitted customer leads are sent, such as receiver name, receiver email, Telegram chat ID, internal dashboard account, or notification routing settings.

**FR-FLR-04:** Company-level users shall not be able to edit, delete, replace, or override the fixed lead receiver.

**FR-FLR-05:** Only the system owner/super admin shall be able to update the fixed lead receiver from a dedicated receiver settings page.

**FR-FLR-06:** When company content or video is changed, the system shall continue sending new leads to the same fixed receiver.

**FR-FLR-07:** The admin dashboard shall display a warning before any super admin changes the fixed receiver, because changing the receiver affects where lead/customer information is delivered.

## 6.8 Admin Dashboard

**FR-AD-01:** The admin dashboard shall be accessible only after successful login.

**FR-AD-02:** The dashboard shall display the following summary statistics on the home screen:

- Total number of active companies.
- Total leads received (all time).
- Leads received today.
- Leads received in the last 7 days.

**FR-AD-03:** The leads view shall display leads in reverse chronological order by default. Admin shall be able to sort by date, company, and name.

**FR-AD-04:** Admin shall be able to filter leads by: company name, date range, and lead status.

**FR-AD-05:** Admin shall be able to search leads by visitor name, phone number, or email address.

**FR-AD-06:** Admin shall be able to export filtered or all leads to a CSV file. The CSV shall include all lead fields and submission timestamp.

**FR-AD-07:** The dashboard shall display a real-time notification badge or indicator when a new lead is received while the admin is logged in.

**FR-AD-08:** The system shall maintain an audit log of all admin actions including: login/logout, company profile create/edit/deactivate, lead export, and notification setting changes. Each log entry shall include: action type, admin user, timestamp, and affected entity identifier.

## 7. Non-Functional Requirements

### 7.1 Performance

**NFR-PERF-01:** The public landing page shall achieve a full page load time of under 3 seconds on a 4G mobile connection (10 Mbps download) under normal server load.

**NFR-PERF-02:** The lead form submission response (acknowledgement to the browser) shall complete within 2 seconds under normal server load.

**NFR-PERF-03:** New leads shall appear in the admin/company rep dashboard within 5 seconds of form submission.

**NFR-PERF-04:** The system shall support at least 100 concurrent visitors across all landing pages without degradation in response time.

**NFR-PERF-05:** Admin dashboard queries (lead list, filter, search) shall return results within 2 seconds for datasets up to 50,000 leads.

### 7.2 Usability

**NFR-USE-01:** The public landing page shall be mobile-first and fully usable on screen widths from 320px to 1,920px.

**NFR-USE-02:** The lead form shall be completable by a first-time visitor with no instructions, as measured by usability testing with 5 test users achieving task completion with no more than 1 error each.

**NFR-USE-03:** The admin dashboard shall display a visible, accessible error message whenever an action fails. Error messages shall be specific (e.g., "Email notification failed — please check SMTP settings") rather than generic (e.g., "An error occurred").

**NFR-USE-04:** All user-facing text shall be available in at least English. Multi-language support is a future feature (see Section 14).

### 7.3 Availability and Reliability

**NFR-AVAIL-01:** The system shall target 99.5% uptime per calendar month, excluding scheduled maintenance windows.

**NFR-AVAIL-02:** Scheduled maintenance windows shall not exceed 2 hours per week and shall be announced to admins at least 24 hours in advance.

**NFR-AVAIL-03:** If the real-time notification service (WebSocket, Firebase, etc.) is unavailable, the landing page and lead form shall continue to function. Leads shall still be saved to the database. The dashboard shall update upon manual refresh.



## 7.4 Scalability

**NFR-SCALE-01:** The system architecture shall support scaling to at least 500 active companies without requiring code changes.

**NFR-SCALE-02:** The lead database shall be designed to support at least 1,000,000 total lead records without query performance degradation below the thresholds in Section 7.1.

## 7.5 Compatibility

**NFR-COMPAT-01:** The public landing page shall function correctly on:

- iOS Safari (latest 2 major versions)
- Android Chrome (latest 2 major versions)
- Desktop Chrome, Firefox, Edge, Safari (latest 2 major versions)

**NFR-COMPAT-02:** The admin dashboard shall function correctly on desktop Chrome, Firefox, and Edge (latest 2 major versions). Mobile support for the admin dashboard is desirable but not required for v1.

## 7.6 Maintainability

**NFR-MAINT-01:** All API endpoints shall be versioned (e.g., /api/v1/...) to allow backward-compatible updates.

**NFR-MAINT-02:** The system shall produce structured application logs (JSON format) at INFO level for all significant events (form submission, notification sent, login, etc.) and ERROR level for all failures.

# 8. Data Requirements

## 8.1 Data Entities

The system shall manage the following core data entities:

- **Company Profile** — as specified in FR-CPM-02.
- **QR Code** — URL, company reference, creation date, status.
- **Landing Page Visit** — timestamp, company, anonymized IP (optional, for analytics).
- **Lead** — all form fields, metadata (timestamp, company, IP hash, status, notification status).
- **User Account** — email, hashed password, role, associated company (for reps), last login, account status.
- **Audit Log** — action, user, timestamp, entity, detail.
- **Notification Log** — lead reference, channel, status, attempt count, last attempt timestamp.

## 8.2 System Limits

| Limit                                    | Value                                 |
|------------------------------------------|---------------------------------------|
| Maximum active companies                 | 500 (configurable by admin)           |
| Maximum leads per company                | Unlimited                             |
| Maximum total leads in system            | 1,000,000 (before archiving required) |
| Maximum company logo file size           | 2 MB                                  |
| Maximum video URL length                 | 2,048 characters                      |
| Maximum company description length       | 500 characters                        |
| Maximum services text length             | 1,000 characters                      |
| Maximum form message length              | 1,000 characters                      |
| Maximum company rep accounts per company | 5                                     |

## 9. Security and Privacy Requirements

### 9.1 Transport Security

**SEC-01:** All data in transit between client and server shall be encrypted using TLS 1.2 or higher. HTTP connections shall be automatically redirected to HTTPS.

**SEC-02:** HTTPS shall be enforced via HSTS headers with a minimum max-age of 31,536,000 seconds (1 year).

### 9.2 Authentication Security

**SEC-03:** All authentication tokens (session cookies or JWTs) shall be transmitted only over HTTPS and shall be marked as HttpOnly and Secure.

**SEC-04:** JWT tokens, if used, shall have a maximum expiry of 24 hours. Refresh tokens shall have a maximum expiry of 7 days.

### 9.3 Access Control

**SEC-05:** The system shall enforce role-based access control (RBAC) at the API level. Client-side access control (hiding UI elements) is not sufficient — server-side checks shall be the authoritative enforcement layer.

**SEC-06:** Company reps shall only be able to access, read, or modify data belonging to their own company. Any attempt to access another company's data shall return a 403 Forbidden response and shall be logged in the audit log.

### 9.4 Input Security

**SEC-07:** All user-supplied input shall be sanitized server-side to prevent SQL injection, XSS, and CSRF attacks.

**SEC-08:** The lead capture form shall include CSRF token protection.

**SEC-09:** The system shall implement rate limiting on the login endpoint: maximum 10 requests per minute per IP address.

**SEC-10:** The system shall implement rate limiting on the lead submission endpoint: maximum 5 submissions per minute per IP address, with a daily cap of 20 submissions per IP per company.

### 9.5 Privacy and Data Protection

**SEC-11:** Visitor lead data (name, phone, email) shall only be accessible to authorized admin users, the fixed lead receiver, and company users explicitly granted access by the system owner. It shall not be shared with or accessible by other companies.

**SEC-12:** The system shall support a data retention policy. By default, lead data shall be retained for 24 months. Admin shall be able to configure the retention period. Leads older than the configured period may be automatically archived or deleted.

**SEC-13:** Visitor IP addresses, if collected, shall be hashed or anonymized before storage. The raw IP address shall not be persisted.

**SEC-14:** The public landing page shall include a link to a privacy notice explaining what data is collected, why, and how it is used. This notice shall be in place before the system goes live.

**SEC-15:** Visitors shall be able to request deletion of their lead data. The system shall provide a mechanism for admin to fulfill such requests (manual deletion from the admin dashboard is acceptable for v1).

## 10. Error Handling and Resilience

**ERR-01 (Form submission — network error):** If the form submission fails due to a network error, the visitor shall see a message ("Something went wrong. Please try again.") and the form shall retain all entered data so the visitor does not need to re-enter it.

**ERR-02 (Form submission — server error):** If the server returns an error during lead saving, the visitor shall see an appropriate error message. The system shall log the error with full detail.

**ERR-03 (Notification failure):** As defined in FR-RLD-06, notification failures shall not prevent lead saving. Admin shall be able to see which leads had failed notifications and re-trigger notifications manually.

**ERR-04 (Video unavailable):** As defined in FR-LP-05, a broken video URL shall not prevent the rest of the landing page from loading.

**ERR-05 (Deactivated company QR scan):** As defined in FR-CPM-05, scanning the QR of a deactivated company shall redirect to a friendly "not available" page, not a 404 error.

**ERR-06 (Database downtime):** If the database is temporarily unavailable at the time of form submission, the system should queue the lead submission for retry. If queuing is not implemented in v1, the visitor shall be shown a clear error message, and the form data shall be preserved for re-submission.

**ERR-07 (Invalid QR URL):** If a visitor scans a QR code that does not match any known company (e.g., corrupted or test QR), the system shall redirect to a branded 404 page explaining the issue.

## 11. Integration Requirements

### 11.1 Email Notifications

**INT-01:** The system shall integrate with an SMTP email service or transactional email API (e.g., SendGrid, Mailgun, or AWS SES) for notification delivery.

**INT-02:** The from-address for notification emails shall be configurable by super admin.

**INT-03:** Notification emails shall be sent to the fixed lead receiver and shall include the full lead details and a direct link to view the lead in the dashboard.

### 11.2 Telegram Notifications (Secondary)

**INT-04:** The system shall support Telegram Bot API integration as an optional notification channel per company.

**INT-05:** Super admin shall be able to configure a Telegram Bot token and the fixed target chat ID for Telegram notification delivery. Company-level users shall not edit this destination.

### 11.3 Video Hosting

**INT-06:** The system shall support video URLs from YouTube and Vimeo as the primary hosted video options. Direct upload to Cloudinary may be supported as an optional enhancement.

**INT-07:** Videos shall be embedded using the hosting provider's iframe embed method. The system shall not download or re-host third-party video content.

## 12. Selected Technology Stack

The system shall be implemented using the following selected MVP technology stack. This stack is chosen because it supports fast development, real-time lead delivery, secure authentication, scalable data storage, and simple deployment.

| Layer / Component | Selected Technology | Purpose |
|-------------------|---------------------|---------|
|-------------------|---------------------|---------|

|                      |                                                          |                                                                                                            |
|----------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Frontend framework   | Next.js with App Router and TypeScript                   | Public landing pages, admin dashboard, routing, SEO-friendly pages, and maintainable application structure |
| Styling and UI       | Tailwind CSS                                             | Mobile-first responsive design, clean admin screens, and fast UI customization per company brand           |
| Backend/API          | Next.js API Routes / Server Actions                      | Lead submission, company profile management, QR generation endpoints, and notification triggers            |
| Database             | Supabase PostgreSQL                                      | Stores company profiles, leads, users, QR links, audit logs, and notification logs                         |
| Authentication       | Supabase Auth                                            | Admin and company representative login, session handling, password reset, and role-based access support    |
| Real-time delivery   | Supabase Realtime                                        | Pushes newly submitted leads to the dashboard within the required real-time threshold                      |
| File storage         | Supabase Storage                                         | Stores company logos and approved media assets                                                             |
| QR generation        | qrcode npm package                                       | Generates downloadable QR code images from each company's permanent landing page URL                       |
| Hosting              | Vercel for the web app and Supabase for backend services | Fast deployment, environment management, and scalable hosting for the MVP                                  |
| Notifications        | Email notification first; Telegram Bot optional          | Sends new lead alerts to the company receiver immediately after successful lead saving                     |
| Form validation      | React Hook Form + Zod                                    | Client-side and server-side validation for lead forms and admin forms                                      |
| Icons and UI helpers | Lucide React                                             | Clean interface icons for dashboard, contact buttons, and landing page components                          |

## 12.1 Recommended MVP Architecture

The first production version should use a single Next.js application connected to Supabase. The public landing page, admin dashboard, API endpoints, authentication, database, storage, and real-time subscriptions shall all work from this unified architecture.

**Selected MVP stack:** Next.js + Tailwind CSS + Supabase PostgreSQL + Supabase Auth + Supabase Realtime + Supabase Storage + qrcode npm + Vercel.

## 12.2 Rationale

This stack avoids unnecessary complexity while still supporting the core business requirement: real-time lead capture and delivery. It also allows the platform to support multiple companies without rebuilding the system for every new client.

## 13. First Version Scope

The following features are mandatory for v1 release:

- Admin login with lockout and password reset
- Add, edit, replace, and deactivate company profiles
- Replaceable video URL and landing page content per company
- Unique permanent QR code per company (PNG download)
- Public company landing page (mobile-first)
- Lead capture form with client-side and server-side validation
- Duplicate submission protection (FR-LCF-07)
- Real-time lead saving to database
- Real-time lead display in admin/company rep dashboard ( $\leq 5$  seconds)
- Email notification for new leads to the fixed receiver with retry on failure
- Telegram notification (optional, if configured)
- Admin lead list with search, filter by company/date, and CSV export
- Role-based access control, including locked receiver configuration
- Session management (timeout, lockout)
- Audit log for admin actions
- HTTPS enforcement
- Deactivated company landing page (friendly message)
- Video fallback if URL is broken
- Mobile-responsive design

## 14. Optional Future Features

Future versions may include:

- Company self-service account registration and management
- QR scan analytics (scan count, device type, geographic data)
- Lead status tracking and sales pipeline view
- Follow-up notes and activity log per lead
- Lead assignment to staff members
- Multi-language support for landing pages
- Multiple landing page templates/themes
- Subscription and payment management
- CRM integration (Salesforce, HubSpot, Pipedrive)
- WhatsApp Business API notification channel
- Push notifications (web push / PWA)
- Bulk QR code printing export
- Two-factor authentication (2FA) for admin accounts
- API access for external integrations

## 15. Acceptance Criteria

The system will be accepted for production release when all of the following criteria are verified:

### Functional Acceptance

| #     | Criterion                                                                                           | Test Method |
|-------|-----------------------------------------------------------------------------------------------------|-------------|
| AC-01 | Each company has a unique QR code that opens the correct landing page                               | Manual test |
| AC-02 | QR code URL does not change after updating/replacing company name, video, or public contact details | Manual test |
| AC-03 | Visitor can watch the video and view company information on mobile and desktop                      | Manual test |

|       |                                                                                                             |                       |
|-------|-------------------------------------------------------------------------------------------------------------|-----------------------|
| AC-04 | Visitor can submit the contact form without creating an account                                             | Manual test           |
| AC-05 | Lead data is saved to the database immediately after form submission                                        | DB verification       |
| AC-06 | Admin/company rep dashboard displays new lead within 5 seconds of submission                                | Automated timing test |
| AC-07 | Email notification is received within 5 seconds of form submission (under normal conditions)                | Timed manual test     |
| AC-08 | Notification failure triggers 3 retries and marks lead with "notification failed" flag                      | Integration test      |
| AC-09 | Company profile update/replacement does not change QR code URL or fixed lead receiver                       | Manual test           |
| AC-10 | Scanning the QR of a deactivated company shows a friendly message, not a 404                                | Manual test           |
| AC-11 | Company rep cannot view another company's leads (RBAC enforcement verified at API level)                    | Automated test        |
| AC-12 | Admin can search, filter, and export leads to CSV                                                           | Manual test           |
| AC-13 | 5 consecutive failed login attempts locks the account for 15 minutes                                        | Manual test           |
| AC-14 | Same visitor/IP is blocked after 3 form submissions in 24 hours                                             | Automated test        |
| AC-15 | A broken video URL does not prevent the landing page from loading                                           | Manual test           |
| AC-16 | Company-level users cannot edit or override the fixed lead receiver                                         | RBAC/API test         |
| AC-17 | After company content/video replacement, a new submitted lead is still delivered to the same fixed receiver | End-to-end test       |

## Non-Functional Acceptance

| #         | Criterion                                          | Threshold                  | Test Method              |
|-----------|----------------------------------------------------|----------------------------|--------------------------|
| NFR-AC-01 | Landing page load time on 4G mobile                | < 3 seconds                | Lighthouse / WebPageTest |
| NFR-AC-02 | Form submission response time                      | < 2 seconds                | Load test                |
| NFR-AC-03 | Dashboard lead update after submission             | ≤ 5 seconds                | Automated test           |
| NFR-AC-04 | HTTPS enforced, HTTP redirects to HTTPS            | 100%                       | Security scan            |
| NFR-AC-05 | All inputs sanitized (no XSS/SQLi vulnerabilities) | 0 critical findings        | OWASP ZAP scan           |
| NFR-AC-06 | Visitor PII not accessible across companies        | 0 cross-company data leaks | Penetration test         |

|           |                                                        |                   |                          |
|-----------|--------------------------------------------------------|-------------------|--------------------------|
| NFR-AC-07 | System handles 100 concurrent visitors without timeout | < 2s avg response | Load test (k6 or JMeter) |
|-----------|--------------------------------------------------------|-------------------|--------------------------|

\*End of SRS v2.1\*